PRACTICAL COURSE WORKBOOK

Data Visualization - Tableau & Python

Turn untidy data into a defensible finding

LEVEL	Intermediate
GUIDED TIME	5 hours across 12 lessons
PROJECT	a cleaned dataset, analysis and decision-ready chart
SKILLS	Tableau, Plotly, Seaborn, Dashboards, Critical evaluation, Documentation
REVIEWED	2026-08-25

WHAT YOU WILL BE ABLE TO DO

- Use Tableau appropriately in a realistic, bounded task.
- Use Plotly appropriately in a realistic, bounded task.
- Use Seaborn appropriately in a realistic, bounded task.
- Use Dashboards appropriately in a realistic, bounded task.

WHO THIS IS FOR

Learners using data and code to answer practical questions.

BEFORE YOU BEGIN

- Basic numeracy and file-management skills

Use this guide beside the interactive lessons. Keep inputs, decisions, tests, limitations and the next improvement.

COURSE MAP

From foundation to finished work

Data Visualization - Tableau & Python is a structured, practical course covering Tableau, Plotly, Seaborn, Dashboards. It emphasizes explainable methods, checked work and a final outcome that can be reviewed.

MODULE	LESSONS AND FOCUS
01 Data foundations	1. Problem framing with Tableau 2. Data types with Plotly 3. Tool setup with Seaborn
02 Analysis	1. Cleaning with Dashboards 2. Exploration with Tableau 3. Modelling with Plotly
03 Evaluation	1. Validation with Seaborn 2. Uncertainty with Dashboards 3. Communication with Tableau
04 Applied project	1. Workflow with Plotly 2. Review with Seaborn 3. Reproducibility with Dashboards

HOW TO STUDY EACH LESSON

1. Read the objective and identify the decision it supports.
2. Reproduce the worked example with the stated input.
3. Test one normal case and one boundary or failure case.
4. Complete the knowledge check without guessing.
5. Save a short evidence note: result, limitation and next action.

CAPSTONE PROJECT

Produce a reviewable Data Visualization - Tableau & Python project using Tableau, Plotly, Seaborn.

DELIVERABLE	A working Data Visualization - Tableau & Python artefact plus an evidence-based self-review.
TOOLS	A suitable Tableau environment, A plain-text decision log, Test data or realistic sample material

Production plan

1. Define the intended user, outcome and constraints.
2. Create the smallest complete result using Tableau.
3. Test one normal case, one boundary case and one failure response.
4. Revise the work from the evidence and preserve before-and-after results.
5. Prepare a concise handover containing method, limitations and next step.

Definition of done

- The output matches the stated outcome.
- Inputs and decisions are reproducible.
- Boundary and failure evidence is included.
- Limitations and responsibility considerations are explicit.

Evidence log

INPUT	Original brief, data, requirements or test material.
DECISIONS	Chosen method and one reasonable alternative.
TESTS	Normal, boundary and failure results.
LIMITATION	What the result does not establish.
REVISION	One evidence-led improvement.

REVIEW AND SOURCES

Make the work credible and repeatable

Final self-review

- Can another learner repeat the method?
- Did I test a difficult case?
- Did I avoid unsupported claims?
- Is the next action proportionate to the remaining risk?

Authoritative references

The Python Tutorial

Python Software Foundation - reviewed 2026-08-21
https://docs.python.org/3/tutorial/

User Guide

scikit-learn - reviewed 2026-08-21
https://scikit-learn.org/stable/user_guide.html

SOURCE NOTE

Product versions, laws and professional standards can change. Recheck the linked primary source before applying version-sensitive information.

NEXT IMPROVEMENT

Choose one weakness found during review and improve it before extending the Tableau scope.